

NSF BIOGRAPHICAL SKETCH

NAME: Powell, Jeaime

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POSITION TITLE & INSTITUTION: Broadening Participation Hackathon Coordinator, XSEDE

(a) PROFESSIONAL PREPARATION -(see PAPPG Chapter II.C.2.f.(a))

INSTITUTION	LOCATION	MAJOR / AREA OF STUDY	DEGREE (if applicable)	YEAR YYYY
Elizabeth City State University	Elizabeth City, NC	Computer Science	BS	2004
Elizabeth City State University	Elizabeth City, NC	Applied Mathematics	MS	2010

(b) APPOINTMENTS -(see PAPPG Chapter II.C.2.f.(b))

2020 - present Broadening Participation Hackathon Coordinator, XSEDE, Austin, TX

2017 - present Lecturer, University of Texas at Austin, Austin, TX

2017 - present Senior Systems Administrator and Technical Research Design Analyst, Texas Advanced Computing Center, Austin, TX

2015 - 2017 Systems Analyst, Cray Incorporated, Orlando, FL

2014 - 2015 Systems Administrator, Old Dominion University, Norfolk, VA

2006 - 2014 Grid Manager and Researcher, Elizabeth City State University, Elizabeth City, NC

2002 - 2007 Technical Training Specialist, Canon Information Technology Services, Chesapeake, VA

(c) PRODUCTS -(see PAPPG Chapter II.C.2.f.(c))

Products Most Closely Related to the Proposed Project

1. Matheny A, Marchetto P, Powell J, Rechner A, Chuah J, McCormick E, Pierce S. LEAF: Logger for ecological and atmospheric factors. HardwareX. 2019 October; 6:e00079-. Available from: <https://linkinghub.elsevier.com/retrieve/pii/S246806721930015X> DOI: 10.1016/j.ohx.2019.e00079
2. Hayden L, Powell J, Akers E. Establishing field and base camp servers for remote sensing of ice sheets in ilulissat, Greenland. 2009 IEEE International Geoscience and Remote Sensing Symposium. 2009 IEEE International Geoscience and Remote Sensing Symposium; ; Cape Town, South Africa. IEEE; c2009. Available from: <http://ieeexplore.ieee.org/document/5418148/> DOI: 10.1109/IGARSS.2009.5418148
3. Cleveland Sean B, Jamthe Anagha, Padhy Smruti, Powell Je'aime, Stubbs Joe, Daniels Michael D, Pierce Suzanne A, Jacobs Gwen A. Tapis-CHORDS Integration: Time-Series Data Support in Science Gateway Infrastructure. Conference: Science Gateways 2019At: San Diego, CA; 2019; c2019.
4. Powell Jeaime, Bailey Hayden Linda, Cannon Amy, Wilson Boyd, Nolte Alexander. Organizing online hackathons for newcomers to a scientific community--Lessons learned from two events. Sixth Annual International Conference on Game Jams, Hackathons, and Game Creation Events; 2021; c2021.

5. Gil Yolanda, Osorio Maximiliano, Ratnakar Varan, Pierce Suzanne, Thorne Nick, Lubbs Peter, others. Towards Democratizing Modeling at Scale. 2021 IEEE 17th International Conference on eScience (eScience); 2021; c2021.

Other Significant Products, Whether or Not Related to the Proposed Project

1. Powell Mr Je'aime, Hayden Linda, Farmer Courtney, McDonald Kalyx. TeraScan Curriculum Development and Integration of SeaSpace Technology into the Classroom. .

(d) SYNERGISTIC ACTIVITIES -(see PAPPG Chapter II.C.2.f.(d))

1. High-performance computing, STEM outreach through education, and remote sensing of environments using edge systems are the passions in my life. The combinations and intermixing of those passions are the fortunate channels through which I have been allowed to utilize my creativity to address opportunities. Throughout my career, I have had the pleasure to be exposed to ideas and technologies early within the development cycle. That exposure has had the long term effect of providing foundational understandings that then allow me to translate that knowledge into methods through which that knowledge can both be applied and taught to the next generation of scientists, researchers, and technology professionals. This journey and love of all things computer began with the gift of a Commodore64 from my father. The ability to create programmatically as well as repair and maintain the technology quickly became a source of joy that was magnified with the addition of a 2400 baud modem and an account on the then "Virginia Pen" bulletin board service in the early-90s. With the world at my fingers, I was able to write high school reports with references from the libraries and schools home to subject matter experts. My world was expanded in the mid-90s with my first two jobs in a research lab and a small computer shop. During the week after school I would learn and apply object-oriented programming techniques on Sun Spark, Next, and Silicon Graphics computer systems at Elizabeth City State University while on the weekends I repaired Windows and Apple System 7 machines for business and home users at a small computer shop named The DataBank. That duality of research-based Computer Science and practical Information Technology skills gave me an understanding of Computer Information Systems cooperatively. My career has been influenced by this intersection of practical and academic formation ideas through my attainment of a Bachelor's of Science in Computer Science and Master's of Science in Applied Mathematics with a concentration in Remote Sensing while also gaining CompTIA certifications in A+, Network+, and Security+.

During my undergraduate degree, I was fortunate to participate in several research programs through Elizabeth City State University, NASA, and the Office of Naval Research. In the summer of 1998, I participated in a program that included teaching at-risk teenage males calculus. During that program, I gained a joy of education and outreach that included a love of teaching to my passions. While attaining my master's I was giving the opportunity to become a research mentor to undergraduate students. This allowed me to work with several groups of undergraduate students to develop, implement, analyze, and publish projects that combined remote sensing techniques with statistical analysis through analysis of variance and neural networks in the Center of Excellence in Remote Sensing Education and Research.

Most recently, I was fortunate to continue my education and outreach work with high school students in the Code@Texas Advanced Computing Center (TACC) program hosted at the University of Texas at Austin. The programs include Robotics (Code@TACC: Robotics), Cyber Security (SecureATX and GenCyber), and Environmental Sensing (DesignSafe and

Code@TACC: Connected). Additionally, I have had the pleasure to teach Introduction to Scientific Programming as a Lecturer and host computer science-themed Hackathons with the Science Gateways Community Institute and the Extreme Science and Engineering Discovery Environment (XSEDE). The penultimate combination of practical research and outreach based education presently exists with my work in the implementation of hackathons. The time-bound events incorporate exposure coupled with mentoring in a format that promotes the practical application of skills towards the resolution of a problem of interest to students. The additional responsibilities added by XSEDE under my position as the Broadening Participation Hackathon Coordinator allows the opportunity to further use that format to expose both students and teachers across fields and levels with the motivation based on topics of concern to the participants.

Research into Information Systems continues to also play a large role in my career. In my role as a Senior Systems Administrator and Technical Research Design Analyst, I have worked at both the edge node and cloud/back-end computing portions of data collection and analysis respectively. As a funded researcher of both the μ PSTREAM project of the PlanetTexas 2050 and Stengl-Wyer grants and ODSACK Cloudify Gateways award I have been allowed the opportunities to create, code, and deploy Internet-of-Things sensors while developing and maintaining the supporting backend infrastructure capture, store, and analyze the data sets. This continuation of a cooperative view of computer information systems has become the current focus of my research and academic expansions in the field.